

A short proof of Hausdorff's theorem on extending metrics

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The aim of this paper is to give a short proof of the following theorem:

THEOREM. *Let X be a metrizable topological space and let $A \subset X$ be a closed set. Then, for every metric ϱ on A which induces the relative topology on A , there is a metric $\bar{\varrho}$ on X which is an extension of ϱ and is compatible with the topology of X . If moreover X is complete-metrizable and ϱ is a complete metric on A , then the extension $\bar{\varrho}$ above can be obtained to be a complete metric on X .*

The first part of the theorem was proved by F. Hausdorff [5] in 1930 (cf. also [6]) and independently by R. H. Bing ([3], Theorem 5) in 1947, whereas the remark concerning the “complete” case was made by P. Bacon [2] in 1968. Let us note that R. Arens ([1], Theorem 3.3) gave in 1952 a relatively simple proof of Hausdorff's theorem; his arguments were based on a close examination of “Dugundji's retraction”.

The proof we are going to present involves (besides other well-known facts) the use of the following lemma of V. L. Klee ([7], pp. 36).

LEMMA. *Let E and F be normed linear spaces and let $K \subset E \times \{0\}$ and $L \subset \{0\} \times F$ be closed subsets of $E \times F$. Then, for every homeomorphism $f: K \xrightarrow{\text{onto}} L$, there is an extension of f to a homeomorphism $\bar{f}: E \times F \xrightarrow{\text{onto}} E \times F$.*

Proof. Denote by p_E and p_F the natural projections of $E \times F$ onto E and F , respectively. Since F is an ANR($\mathfrak{M}$) ([4], Theorem 4.1), the function $p_F \circ f: K \rightarrow F$ can be extended to $\lambda: E \times \{0\} \rightarrow F$. We put $f_1(\alpha, \beta) = (\alpha, \beta + \lambda(\alpha, 0))$, $(\alpha, \beta) \in E \times F$; f_1 is then a homeomorphism of $E \times F$ onto itself satisfying $f_1(\alpha, \beta) = (\alpha, p_F \circ f(\alpha, \beta))$ for $(\alpha, \beta) \in K$. Similarly, there is a homeomorphism $f_2: E \times F \xrightarrow{\text{onto}} E \times F$ such that $f_2(\alpha, \beta) = (p_E \circ f^{-1}(\alpha, \beta), \beta)$ for $(\alpha, \beta) \in L$. We then have $f_2 \circ f(\alpha, \beta) = f_1(\alpha, \beta)$ for $(\alpha, \beta) \in K$, whence $\bar{f} = f_2^{-1} \circ f_1$ is the desired extension of f .

Now we pass to the proof of the theorem; Bacon's remark will be considered in brackets.

Proof of the theorem. By [8], (A, ϱ) can be isometrically imbedded as a closed subset of a [complete] normed linear space $(F, \|\cdot\|_F)$; let $h_0: A \rightarrow F$ be the embedding.⁽¹⁾ Similarly, there is a topological embedding $g_0: X \rightarrow E$ of X onto a closed subset of a [complete] normed linear space $(E, \|\cdot\|_E)$. We put $\|(\alpha, \beta)\| = \|\alpha\|_E + \|\beta\|_F$ for $(\alpha, \beta) \in E \times F$ and we set $g = g_0 \times 0: X \rightarrow E \times \{0\}$, $h = 0 \times h_0: A \rightarrow \{0\} \times F$, $K = g(A)$ and $L = h(A)$. By the Lemma, the homeomorphism $f = h \circ g^{-1}: K \xrightarrow{\text{onto}} L$ can be extended to a homeomorphism $\bar{f}: E \times F \xrightarrow{\text{onto}} E \times F$. Obviously, $u = \bar{f} \circ g$ is a topological embedding of X onto a closed subset of $E \times F$ and, since $h(A) \subset \{0\} \times F$, the restriction $u|_A = h$ is an isometric embedding of (A, ϱ) into the [complete] normed linear space $(E \times F, \|\cdot\|)$. The metric

$$\bar{\varrho}(x_1, x_2) = \|u(x_1) - u(x_2)\|, \quad x_1, x_2 \in X,$$

is the extension of ϱ we have been looking for.

References

- [1] R. Arens, *Extension of functions on fully normal spaces*, Pacific Journ. of Math. 2 (1952), pp. 11–22.
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- [3] R. H. Bing, *Extending a metric*, Duke Math. J. 14 (1947), pp. 511–519.
- [4] J. Dugundji, *An extension of Tietze's theorem*, Pacific Journ. of Math. 1 (1951), pp. 353–367.

⁽¹⁾ **Added in proof.** For the sake of completeness we insert here a construction of h_0 (it seems to be somewhat simpler than that given by E. Michael [8]). Fix $p \in A$, denote by Z the set of all finite subsets of A , and let E be the Banach space of bounded real functions on Z (with the supremum norm). Define $h_0: A \rightarrow E$ by

$$h_0(x) = f_x \in E, \quad \text{where} \quad f_x(S) = \text{dist}_\varrho(x, S) - \text{dist}_\varrho(p, S) \text{ for } S \in Z.$$

Given $a, b \in A$ and $S \in Z$ we have $|(f_a - f_b)(S)| \leq \varrho(a, b) = |(f_a - f_b)(\{b\})|$, what implies $\|f_a\| \leq \varrho(p, a)$ and $\|f_a - f_b\| = \varrho(a, b)$. Therefore it remains to show that h_0 is closed when considered as a map into $F =$ the linear span of $h_0(A)$. This is because if $(x_n) \in A^\infty$ is a sequence such that $\lim_n h_0(x_n) = \sum_{i=1}^m \lambda_i f_{a_i}$ for some $\lambda_1, \dots, \lambda_m \in R$ and $a_1, \dots, a_m \in A$, then setting $S = \{a_1, \dots, a_m, p\}$ we get $\lim_n \text{dist}_\varrho(x_n, S) = \lim_n f_{x_n}(S) = (\sum_{i=1}^m \lambda_i f_{a_i})(S) = 0$; thus (x_n) must contain a subsequence converging to a point of $\{a_1, \dots, a_m, p\}$. Our argument shows also that $h_0(A \setminus \{p\})$ is a linearly independent set (observe that $h_0(x) \in \text{span}\{h_0(a_i) : i \leq m\}$ implies $x \in \{a_1, \dots, a_m, p\}$). The embedding constructed is very similar to the classical embedding of Kuratowski-Kunugui, see Fund. Math. 25 (1935), p. 543, and Proc. Imp. Acad. Japan 11 (1935), p. 351.

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- [8] E. Michael, *A short proof of the Arens-Eells embedding theorem*, Proc. Amer. Math. Soc. 15 (1964), pp. 415–416.

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